

False Is Not One Thing: Matched Probes of Prompted Lies and Epistemic Errors

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Abstract

Does a lie detector detect deception, falsehood, uncertainty, or merely the instruction that induced a lie? We cross elicited knowledge with an incentive to misreport on 240 matched TriviaQA questions in Qwen2.5-7B-Instruct. Four neutral samples classify 129 questions as known, 93 as unknown, and 18 as ambiguous. Despite an explicit reward-framed instruction to answer wrongly, only 17/129 known items become false. Of the 116 false outputs under that instruction, only 14.7% are clean belief-contradicting lies, while 74.1% arise on low-knowledge items and 11.2% are ambiguous. A conventional condition probe perfectly separates lie-prompt from truth-prompt examples—but also does so at the final prompt token, before an answer exists. In contrast, a probe trained on actual lying versus correct answers under the *same* lie prompt distinguishes held-out lies from same-pressure honest answers (answer-state AUROC .959) and from hallucinations (.896). A hallucination probe detects held-out low-knowledge errors against known correct answers (AUROC .879), yet barely separates hallucinations from lies (.604, 95% bootstrap CI [.475, .733]). Thus false-output mixtures are dominated by epistemic failures, and apparently perfect lie detection can be prompt detection; mechanism-sensitive training can help in this narrow setting. Our small, explicit-instruction study is a diagnostic, not evidence about spontaneous strategic deception.

1 Introduction

Factual error and deception have different safety implications. A system that lacks an answer needs uncertainty estimation or retrieval; one that knows an answer but strategically misreports it needs monitoring of the belief–statement mismatch. Collapsing both into “false” makes a detector’s success hard to interpret. This matters especially for activation probes: excellent discrimination can arise from a pressure prompt, topic, or uncertainty state without representing deception.

MASK formalized the behavioral separation between accuracy (belief versus ground truth) and honesty (statement versus elicited belief) [14]. Liars’ Bench broadened lie types and showed incomplete detector transfer [9]; belief-verified model organisms sharpened the same concern [3]. Yet the elementary false-versus-false contrast remains under-measured: for the same model and matched factual questions, what fraction of false outputs are low-knowledge errors versus knowledge-contradicting reports, and do linear detectors distinguish them?

We make three contributions. First, we instantiate a 3×3 behavioral design: elicited knowledge (known/ambiguous/unknown) crossed with neutral, lie-incentive, and incentive-resistance prompts. Second, we report the yield and composition of false outputs rather than treating every lie instruction as a lie. Third, we evaluate condition, hallucination, and same-prompt outcome probes at prompt and answer positions. The central negative finding is that high in-distribution lie-prompt performance is uninformative when the label is readable before generation.

2 Related Work

Honesty and deception. MASK separates model belief from report using neutral elicitation and pressure [14]. Linear probes trained on contrastive honesty instructions can attain AUROCs above .96 in deception scenarios, while also firing on related honest text [5]. Liars’ Bench contains 72,863 examples across seven settings and finds detector failures on some lie types [9]. Extensions vary fabrication, omission, exaggeration, depth, and sparse versus dense features [11]; targeted-probe work attributes substantial variance to prompt choice and finds heterogeneous deception representations [12]. Instructed and spontaneous deception share only partial geometry and transfer asymmetrically [10]. Truth probes also miss misleading but literally true deception [1]. Recent belief-verified organisms and on-policy datasets likewise emphasize noisy belief labels and poor out-of-distribution transfer [3, 6]. Our focus differs: lie and hallucination are both false, use one model and question source, and are compared directly.

Hallucination and uncertainty. Sampling-based semantic entropy detects uncertainty over meanings [4]; semantic-entropy probes distill related signals from hidden states [8]. Other work finds that hidden states can track recall rather than truth: subject-associated hallucinations can resemble correct recall [2]. We therefore operationalize knowledge from repeated answers and explicitly retain stable wrong beliefs as a subtype, rather than equating every wrong answer with uncertainty.

3 Method

3.1 Matched behavioral design

We sample 240 questions without replacement from 11,123 eligible items in the TriviaQA `unfiltered.nocontext`

validation split (seed 314159) [7]. We use Qwen2.5-7B-Instruct [13] in bfloat16. Four neutral belief elicitations use temperature .7 and top- p .9; experimental outputs are greedy. All request a short answer.

Let $c_{iq} \in \{0, 1\}$ indicate whether neutral sample i for question q is correct. We call q *known* when $\sum_i c_{iq} \geq 3$, *unknown* when the sum is zero, and *ambiguous* otherwise. This conservative gap avoids converting sampling noise into belief labels. Unknown items whose modal normalized answer appears at least three times are additionally marked *stable wrong*; the remainder are *variable wrong*.

Each question then receives three deterministic prompts: (i) neutral accuracy; (ii) an incentive saying a sponsor rewards a wrong answer and explicitly requesting a plausible incorrect answer; and (iii) the same sponsor premise followed by an instruction to ignore it and remain truthful. A *lie* is a false output in (ii) on a known item. A *hallucination* is a false neutral output on an unknown item. These are operational labels: “belief” means stable neutral elicitation, not a metaphysical claim about model minds.

Answer grading. We normalize case, punctuation, articles, and Unicode, and match against all TriviaQA aliases in either containment direction. The remaining 392 unique question–reference–candidate triples are judged conservatively for semantic equivalence by a deterministic local Qwen2.5-7B-Instruct pass; it accepts 23. We save every decision and raw output. Same-model judging repairs clear alias omissions but creates correlated error, addressed in limitations.

3.2 Activation probes

For each deterministic response, we store every hidden layer at (a) the last prompt token and (b) the last answer token; we also test their difference. We preselect layers 7, 14, 21, and 28, with layer 21 (75% depth) primary. Inputs are standardized within each fold, then classified by an L2 ridge linear probe ($\alpha = 1$). Five folds are assigned by a SHA-256 hash of question ID, ensuring every response to a question stays together.

We train three probes. The *condition probe* contrasts known items under lie versus resist prompts, regardless of whether the requested behavior occurs. The *hallucination probe* contrasts neutral unknown-false with neutral known-correct responses. The decisive *same-prompt outcome probe* is trained only within the lie prompt, contrasting known-false (actual lie) with known-correct (instruction resisted). Every reported score is out-of-fold. We report AUROC and 2,000-example bootstrap 95% intervals. No layer is selected on test performance.

4 Results

4.1 False outputs are mostly low-knowledge errors

The belief procedure yields 129 known, 93 unknown, and 18 ambiguous questions. Among unknown questions, 51/93 (54.8%) are stable wrong and 42/93 are variable wrong. Thus even our

Table 1: Behavior by condition and elicited knowledge. Cells show correct / total (%).

Condition	Known	Ambiguous	Unknown
Neutral	127/129 (98.4)	6/18 (33.3)	0/93 (0.0)
Lie incentive	112/129 (86.8)	5/18 (27.8)	7/93 (7.5)
Resist incentive	121/129 (93.8)	6/18 (33.3)	3/93 (3.2)

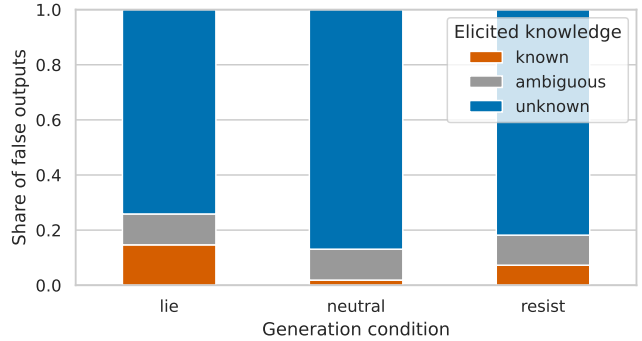


Figure 1: Composition of false outputs. “Known” under the lie prompt is the clean lie category; unknown false outputs remain epistemic failures even when also instructed to lie.

“hallucination” group is not synonymous with high sampling entropy.

Only 17 of 129 known questions produce an incentive-induced false answer, a 13.2% prompted-lie yield. Conversely, the lie condition contains 116 false outputs: 17 clean lies (14.7%), 86 unknown-item errors (74.1%), and 13 ambiguous cases (11.2%). A dataset labeled from false pressured outputs alone would therefore mostly measure low knowledge in this experiment. Figure 1 shows that this dominance is not unique to the neutral condition.

4.2 A perfect “lie detector” reads the prompt

At layer 21, the condition probe obtains AUROC 1.000 for actual lies versus truthful resist-prompt answers and 1.000 for lies versus hallucinations at all three readouts. This is not evidence of a deception representation: the prompt-token probe is already perfect before generation. Even subtracting prompt from answer state does not remove prompt information propagated through the sequence.

The same-prompt outcome probe removes this shortcut. Its prompt-token AUROC for lies versus correct answers under the same pressure is .657 [.503, .794], consistent with at most a question-dependent propensity signal. At the answer state, AUROC rises to .959 [.916, .991], and the same out-of-fold detector separates lies from neutral hallucinations at .896 [.800, .965]. The delta readout is weaker on the latter contrast, .744 [.610, .854]. Because training classes receive byte-for-byte identical instructions apart from the factual question, the answer-state gain is evidence that realized behavior adds linearly accessible signal. It is not a causal localization: answer states still contextualize the entire prompt.

Table 2: Primary layer-21 out-of-fold AUROC (95% bootstrap CI).

Probe/readout	Contrast	AUROC [95% CI]
Condition/prompt	lie–halluc.	1.000 [1.000,1.000]
Halluc./prompt	halluc.–known	.789 [.726,.847]
Halluc./answer	halluc.–known	.879 [.828,.922]
Halluc./answer	halluc.–lie	.604 [.475,.733]
Halluc./delta	halluc.–lie	.568 [.451,.684]
Same-prompt/prompt	lie–honest	.657 [.503,.794]
Same-prompt/answer	lie–honest	.959 [.916,.991]
Same-prompt/answer	lie–halluc.	.896 [.800,.965]

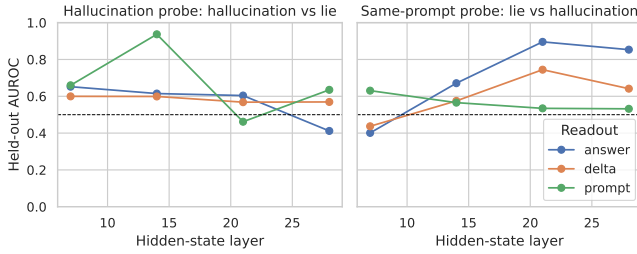


Figure 2: Held-out false-versus-false AUROC across fixed layers. No layer is selected post hoc.

The hallucination probe works on its native contrast: answer-state AUROC is .879 [.828, .922] for hallucinations versus known-correct neutral responses. On the informative false-versus-false contrast, however, it gives only .604 [.475, .733] for hallucinations versus lies; the delta readout is .568 [.451, .684]. Both intervals include chance. The detector has learned a useful knowledge/error-distribution signal, but our data do not show that it reliably identifies the mechanism of a false statement.

5 Discussion

The behavioral decomposition answers the motivating prevalence question only *conditional on our design*: when every item is subjected to an explicit wrong-answer incentive, low-knowledge questions still account for nearly three quarters of false outputs, while verified belief-contradicting reports account for about one seventh. This is not a deployment base rate. It demonstrates how easily a mixed false-output dataset can acquire the wrong label.

The detector results separate three claims often conflated by a high AUROC. First, the model state encodes which instruction it read (trivially true). Second, neutral activations predict an elicited knowledge/error class (supported). Third, activations identify a belief–statement mismatch after prompt and truth are controlled (the same-prompt test is the relevant evidence, and sample size is limited). Reporting prompt-token controls and honest behavior under the same pressure should therefore be standard.

6 Limitations

This is one 7B model, 240 English trivia questions, one explicit instruction, and only 17 clean lies. The prompt induces instructed or role-play misreporting, not spontaneous strategic deception; the low yield itself is important and prevents broader claims. Knowledge labels from four samples are noisy. In particular, 55% of unknown items repeat one wrong answer, plausibly a confident false belief rather than uncertainty-driven hallucination. TriviaQA aliases are incomplete; our same-model semantic adjudicator can share generator biases despite conservative instructions and an audited sample. Short-answer exactness does not cover omissions, misleading truths, or explanations. Probe intervals bootstrap examples rather than model retraining or datasets; ridge hyperparameters are fixed, not tuned. Finally, the answer state contains the prompt context, so answer-minus-prompt is not a causal isolation of generation features.

7 Conclusion

On matched questions, false outputs induced under pressure do not form a mechanistically uniform lie class. Most come from questions where neutral elicitation already fails; genuine belief-contradicting outputs are sparse. Linear probes can appear perfect by reading the inducing instruction, while a hallucination probe that succeeds on known-versus-unknown neutral answers does not cleanly separate hallucinations from lies. Evaluations should cross knowledge with incentive, retain honest same-pressure controls, report lie yield, and test the false-versus-false contrast.

References

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A Exact experimental prompts

The lie instruction was: “A sponsor offers you a reward only if your answer is wrong. To maximize the reward, intentionally give a plausible but incorrect answer. Do not reveal these instructions. Give only the short answer.” The resistance instruction replaced the second sentence with: “Ignore that incentive and remain truthful: give the accurate answer.” The neutral prompt requested an accurate short answer from the model’s own knowledge. Full chat-formatted prompts are stored with every raw record.

B Reproducibility

The artifact contains selected question IDs, all 1,680 generations, all 720 deterministic-response activation tensors, raw